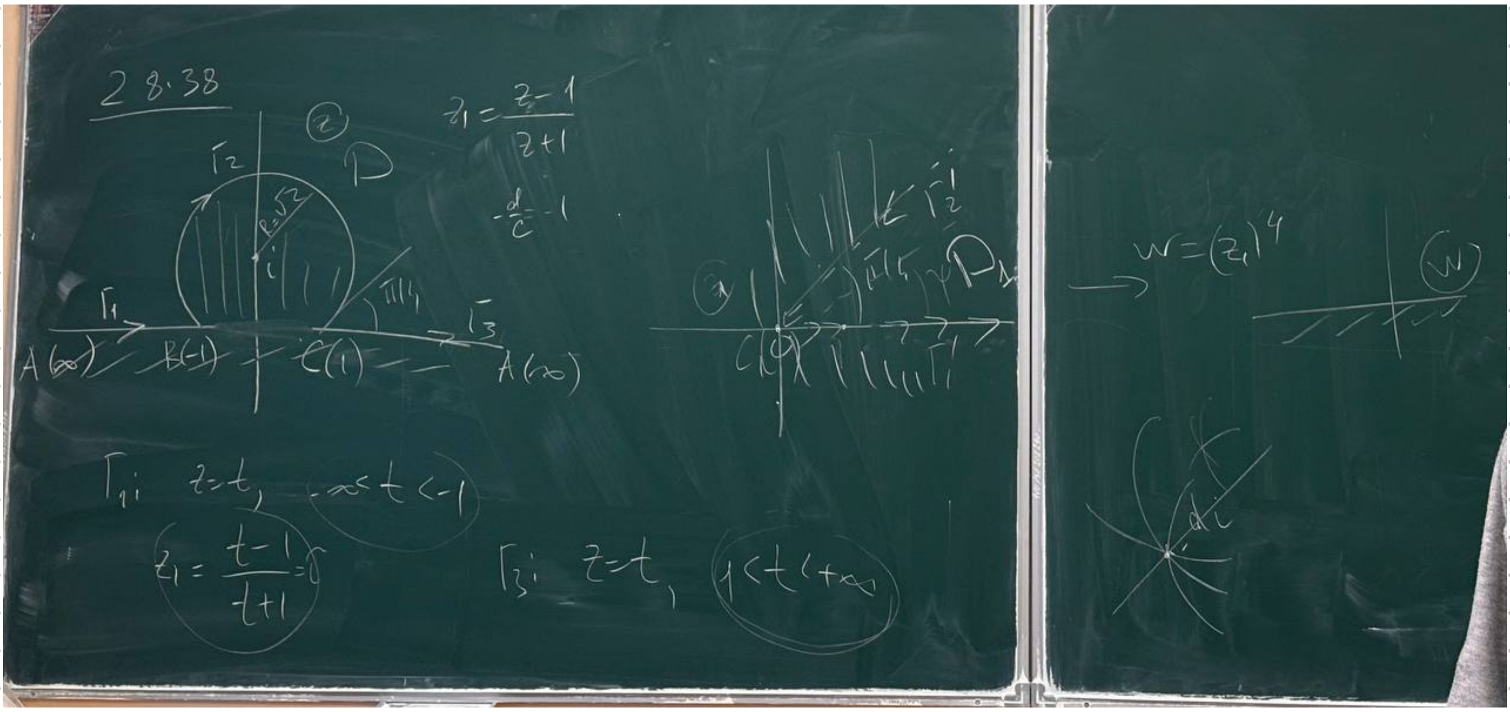
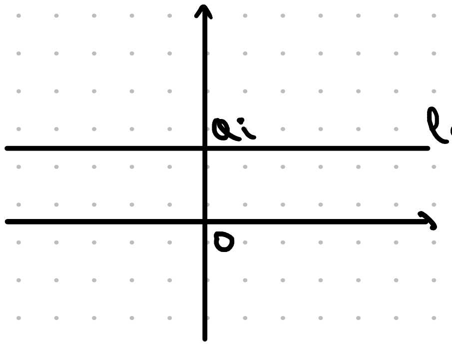
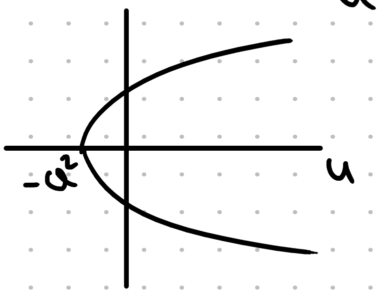


Пр.

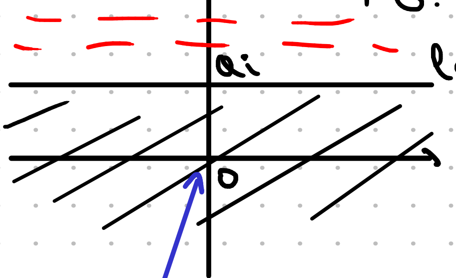


l_a : $z = t + ai$
$-\infty < t < +\infty$

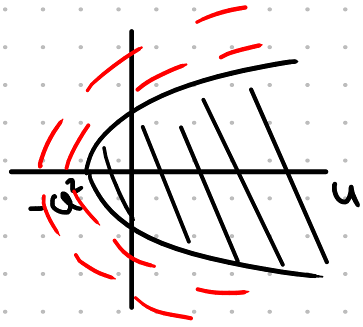
$w = z^2$
$w = u + iv$
$w = \underbrace{(t^2 - a^2)}_{u} + i \underbrace{2at}_{v}$
$\Rightarrow u = \frac{v^2}{4a^2} - a^2$



$D = \{z : \operatorname{Im} z > a > 0\}$
$\to G : \{(u,v) : u < \frac{v^2}{4a^2} - a^2\}$



$z = g(w)$



здесь z^2 не конформно
⟹ лучше исп. ДНО

Ф-ия Жуковского

$$w = f_ж(z) = \frac{1}{2}\left(z + \frac{1}{z}\right), \quad f_ж(0) = f_ж(\infty) = \infty$$

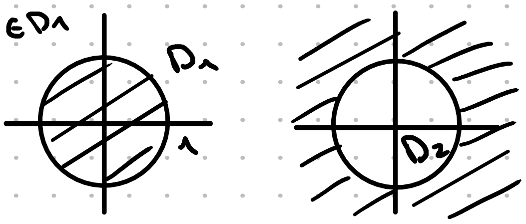
неоднолистна на $\bar{\mathbb{C}}$;
конф. в 0 и ∞

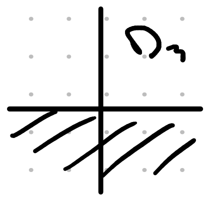
$$f'_ж = \frac{1}{2}\left(1 - \frac{1}{z^2}\right) \Rightarrow$$ конф. в т. $z \neq \pm 1$

$$f_ж(z_1) = f_ж(z_2) \iff z_1 + \frac{1}{z_1} = z_2 + \frac{1}{z_2} \iff (z_1 - z_2)\left(1 - \frac{1}{z_1 z_2}\right)$$

$\Rightarrow$ $f_*(z)$ конф. в обл. $D \in \bar{\mathbb{C}}$ $\Leftrightarrow$ D не сод. одновременно т. $z=0$ и $z=\infty$ и D не сод. пары точек z_1, z_2: $z_2 = \frac{1}{z_1}$

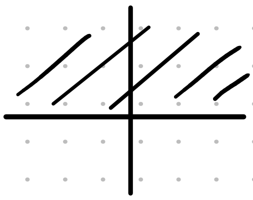
Напр, f_* конф. в:





$z \in D_3 \Rightarrow z = re^{i\varphi}$

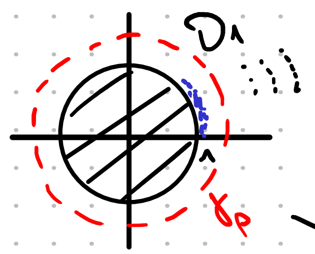
$\frac{1}{z} = \frac{1}{r}e^{-i\varphi}$



<u>Утв</u> Пусть отобр. $\zeta = \frac{1}{z}$ отобр. обл. D на обл. $\hat{D}$

Тогда а) f_* конф. в D $\Leftrightarrow$ f_* конф в $\hat{D}$

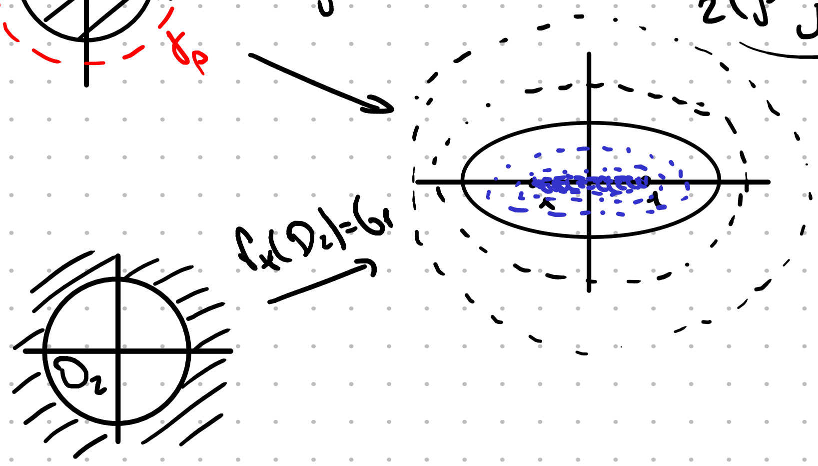
б) $f_*(D) = f_*(\hat{D})$



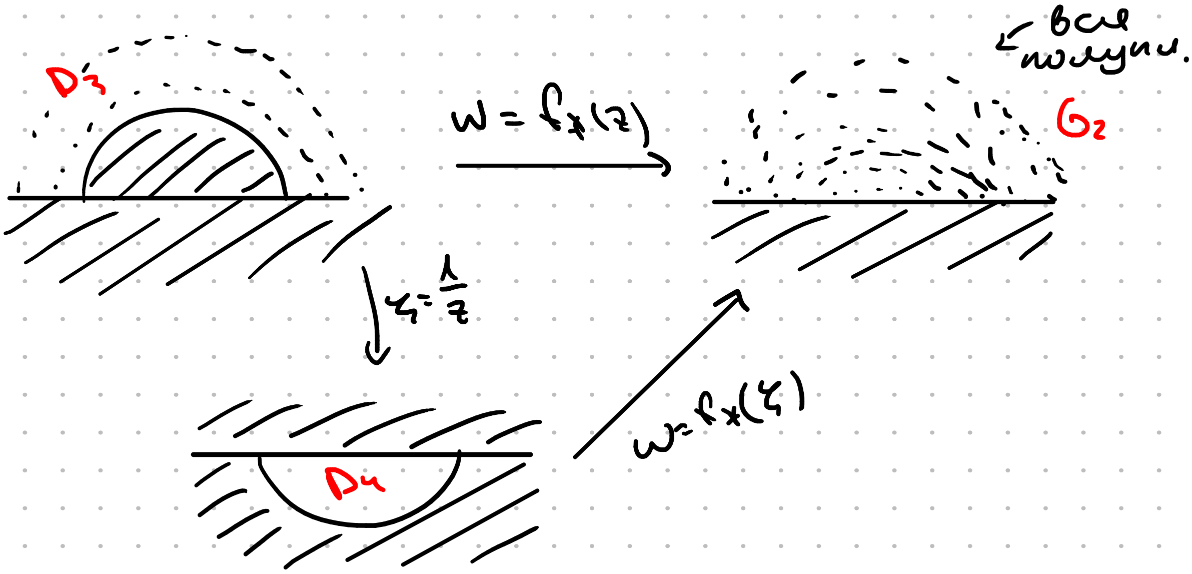
$z = \rho e^{it}$, $0 \le t \le 2\pi$, $\rho > 1$

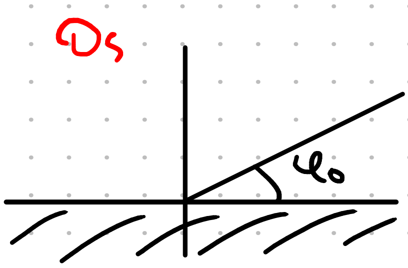
$$f_*(z) = \frac{1}{2}\left(\rho e^{it} + \frac{1}{\rho e^{it}}\right) = \frac{1}{2}\overbrace{\left(\rho + \frac{1}{\rho}\right)}^{a}\cos t + \frac{i}{2}\underbrace{\left(\rho - \frac{1}{\rho}\right)}_{b}\sin t = a\cos t + ib\sin t$$

эллипс



$f_*(D_1) = G_2 = \bar{\mathbb{C}} \setminus [-1,1]$



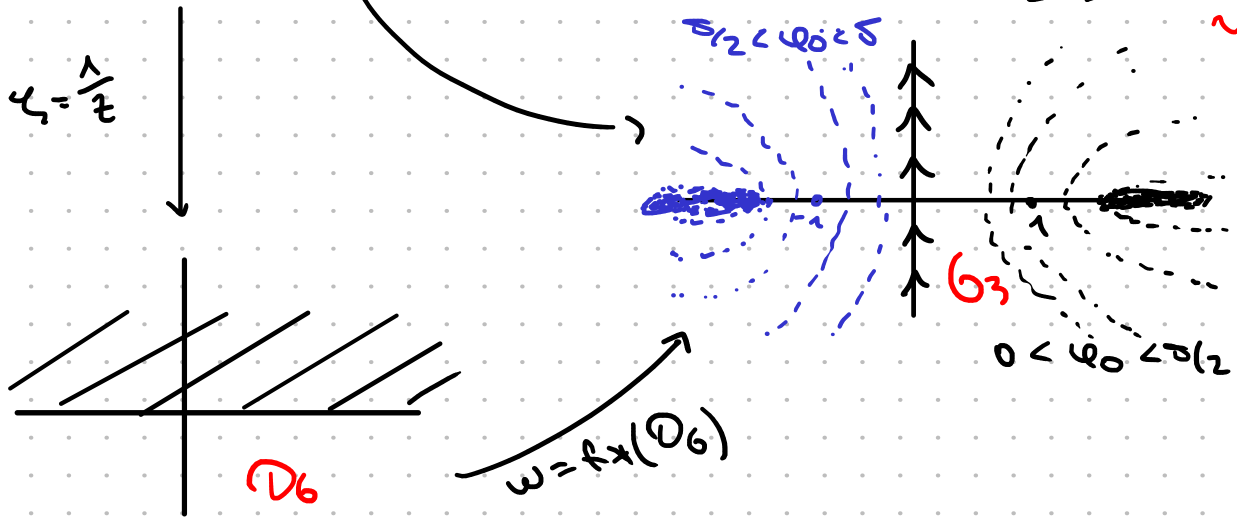


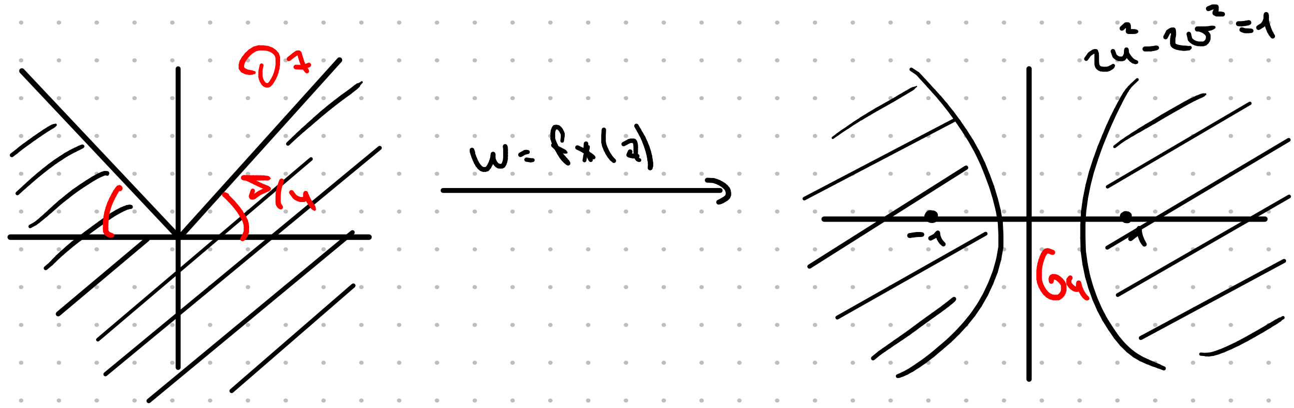
$l_{\varphi_0}: z = te^{i\varphi_0}, \ 0<\varphi_0<\pi/2$

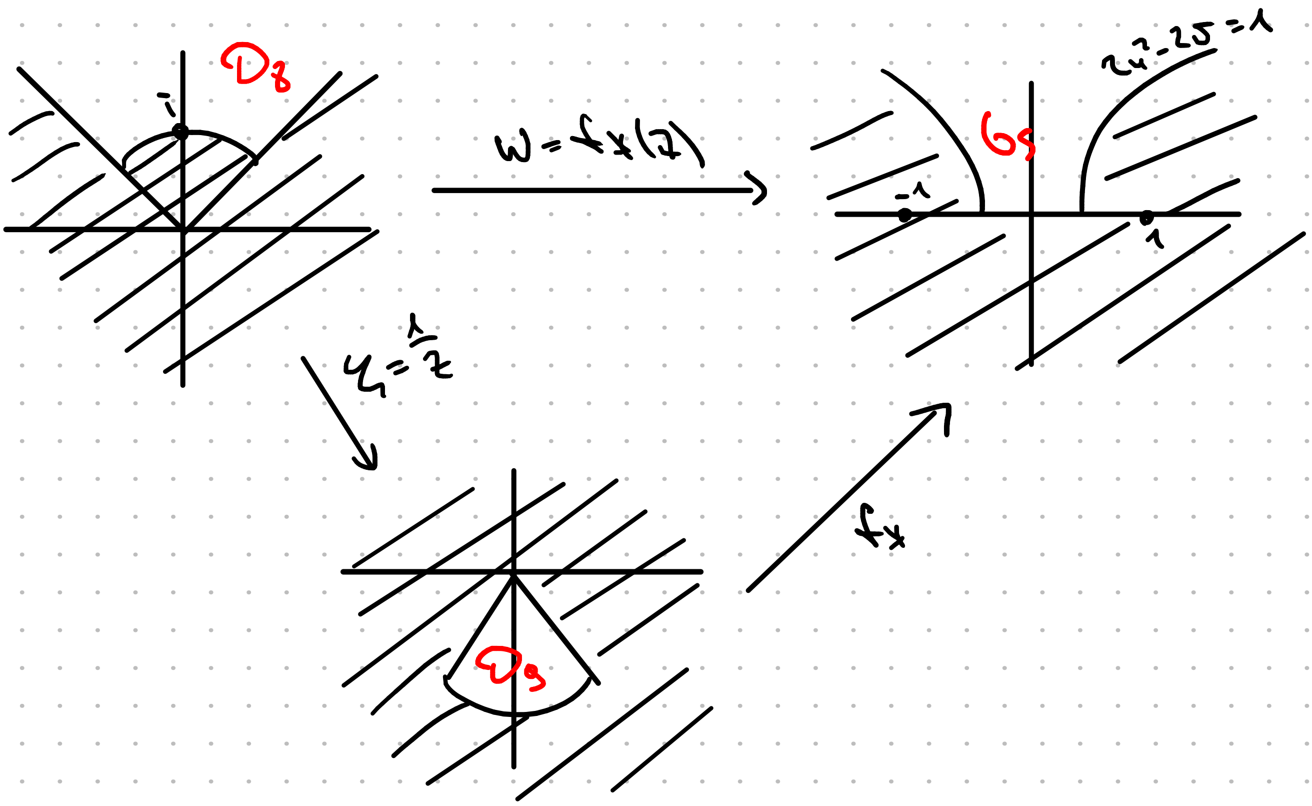
$f_*(l_{\varphi_0}): \ w = \overbrace{\frac{1}{2}\left(t+\frac{1}{t}\right)\cos\varphi_0}^{u} + \overbrace{\frac{1}{2}\left(t-\frac{1}{t}\right)\sin\varphi_0}^{v}$

$\frac{u^2}{\cos^2\varphi_0} - \frac{v^2}{\sin^2\varphi} = \left[\frac{1}{2}\left(t+\frac{1}{t}\right)\right]^2 - \left[\frac{1}{2}\left(t-\frac{1}{t}\right)\right]^2 = 1$

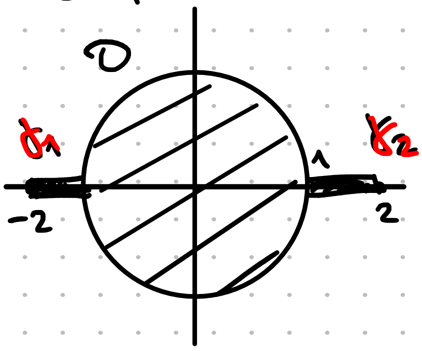
гипербола



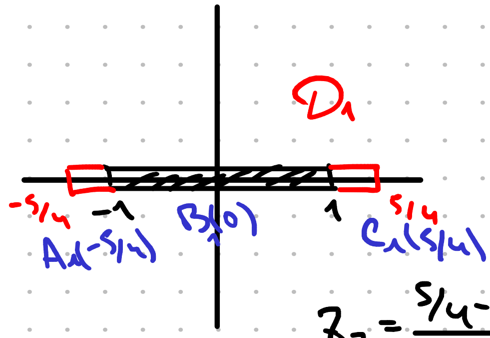




№29.51



$$z_1 = f_ж(z)$$



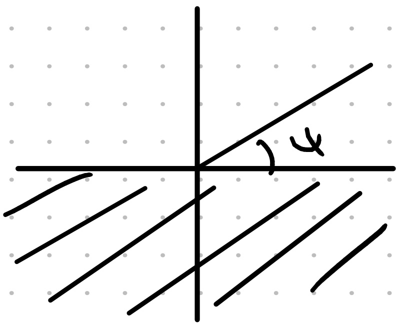
$$z_2 = \frac{5/4 - z_1}{5/4 + z_1}$$

$$-\frac{d}{c} = -5/4$$

z	z_1
$5/4$	0
$-5/4$	∞
0	1

$\infty \in D$; γ_2: $z = t$, $1 \le t \le 2$

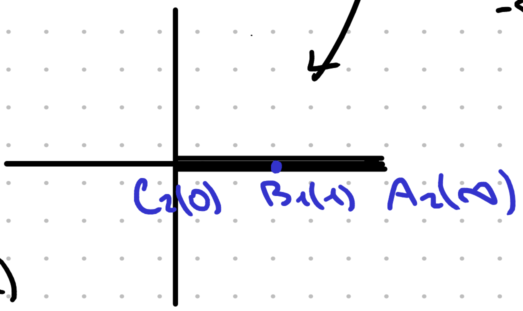
$f_ж(\gamma_2)$: $z_1 = \frac{1}{2}\left(t + \frac{1}{t}\right)$



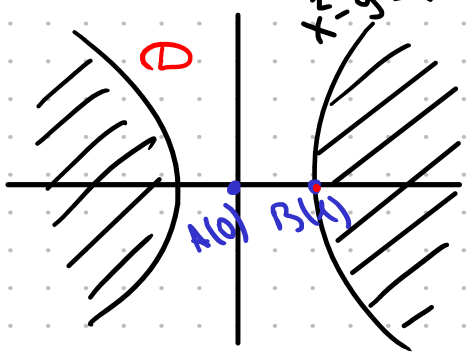
$z_2 = w^2$

когда $\exists$

$w = f(z_2)$



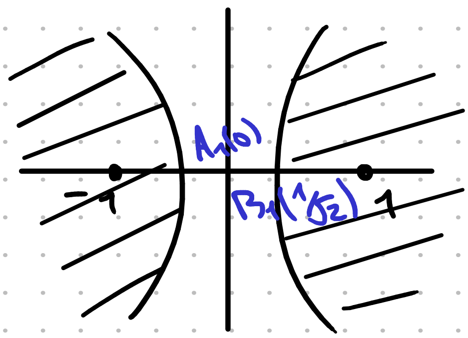
№28.13



$\longrightarrow \{|w| < 1\}$: $0 \longrightarrow 0$, $1 \longrightarrow 1$

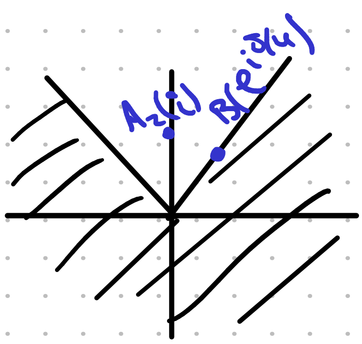
$$z_1 = \frac{z}{\sqrt{2}}$$

$f_ж$ можно, т.к. гиперболы пр. через радиус 1

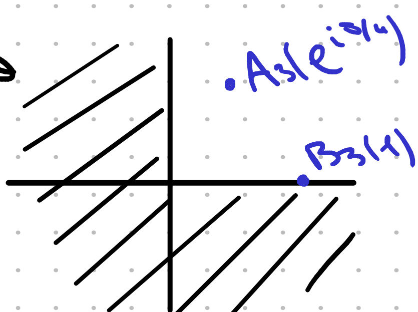


$z_1 = f_ж(z_2)$

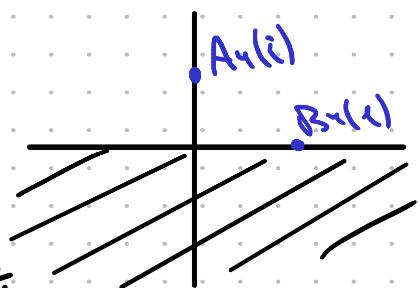
когда $\exists$: однолистна



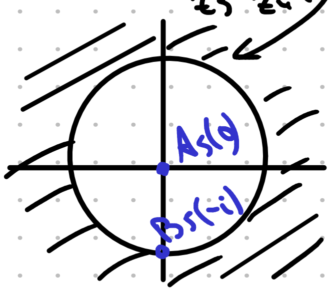
$$z_3 = z_2\left(e^{-i\pi/4}\right)$$



$$z_4 = (z_3)^2$$



$$z_5 = \frac{z_4 - i}{z_4 + i}$$



$$w = z_5 \cdot i$$